Project: Worked examples
Title: Rolling bearing life (ISO 281)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

The basic and modified rating life of a rolling bearing under a steady load or a duty cycle, at a chosen reliability, against a required life; and the static safety factor against a required value.

WHAT IT CALCULATES	RELATION
Equivalent load	$P = X_1 \cdot F_r + Y_1 \cdot F_a$ when $F_a/F_r \le e_{lim}$ ($X_1 = 1$, $Y_1 = 0$ for a single-row radial ball bearing), else $P = X \cdot F_r + Y \cdot F_a$ ($Y = 0$ is refused when an axial load acts above e_{lim})
Duty cycle	$P_{eq} = (\Sigma\ P_i\text{^}p \cdot U_i\ /\ \Sigma\ U_i)\text{^}(1/p)$
Basic rating life	L_{10} = (C/P)^p million revolutions, p = 3 ball, 10/3 roller
Life in hours	$L_{10h} = 10^6 \cdot L_{10}\ /\ (60 \cdot n)$
Reliability factor (ISO 281:2007, derived)	$a_1 = 0.95 \cdot [\ln(100/R)/\ln(100/90)]\text{^}(2/3) + 0.05$
Modified life	$L_{nm} = a_1 \cdot a_{ISO} \cdot L_{10}$
Static safety	$P_0 = \max(X_0 \cdot F_r + Y_0 \cdot F_a, F_r)$, $s_0 = C_0/P_0$

Basis: Dynamic equivalent load and basic rating life[1](cl. 6–7); Modified rating life[1](cl. 9); Static equivalent load and static safety factor[2](cl. 6–7); Rolling-contact bearings[3](11th ed., Ch. 11).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Bearing type	ball (p = 3)		assumption: Deep groove ball bearing.	any
F_r	Radial load	2.5	kN	assumption: Radial reaction from the shaft calculation.	0 … 2000 kN
F_a	Axial load	0	kN	assumption: Locating bearing elsewhere; no axial load here.	0 … 2000 kN
n	Speed	1500	rpm	spec: Pump datasheet, rated speed	1 … 50000 rpm
C	Basic dynamic load rating	20.3	kN	spec: SKF catalogue, 6206	0.1 … 10000 kN
C_0	Basic static load rating	11.2	kN	spec: SKF catalogue, 6206	0.1 … 10000 kN
e_{lim}	Load ratio limit e	0.22		spec: Manufacturer factor for single-row deep groove bearings	0 … 5

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
X_1	Radial factor X_1 for $F_a/F_r \le e$ (1 for a single-row radial ball bearing)	1		default	0 … 5
Y_1	Axial factor Y_1 for $F_a/F_r \le e$ (0 for a single-row radial ball bearing; above 0 for double-row and paired bearings)	0		default	0 … 10
X	Radial factor X for $F_a/F_r > e$	0.56		spec: Manufacturer factor for single-row deep groove bearings	0 … 5
Y	Axial factor Y for $F_a/F_r > e$ (from the manufacturer; Y = 0 makes the axial load count for nothing and is refused when one acts)	2		spec: Manufacturer factor for single-row deep groove bearings	0 … 10
X_0	Static radial factor X_o	0.6		spec: Manufacturer static factor	0 … 5
Y_0	Static axial factor Y_o	0.5		spec: Manufacturer static factor	0 … 10
R	Reliability	95	%	assumption: Client reliability target.	90 … 99.95 %
a_{iso}	Life modification factor a_{ISO} (from the manufacturer's tool)	2		assumption: Read from the manufacturer's life tool for clean ISO VG 68 oil.	0.1 … 50
L_{req}	Required life	5000	h	spec: Pump specification, minimum bearing life	1 … 1000000 h
$s_{0,req}$	Required static safety factor	1		assumption: Normal running, smooth loads.	0.5 … 4

3. Assumptions

A1. Ratings C and C_o, factors X, Y, e, X_o, Y_o and a_{ISO} are the manufacturer's values for the bearing, entered as reference records. (No catalogue table ships with TempestOS.)
A2. a_1 uses the ISO 281:2007 Weibull form (0.64 at 95 %, 0.25 at 99 %); ISO 281:1990 gave 0.62 and 0.21. (Edition stated on the sheet.)
A3. Loads are steady within each duty step; speed is constant. (Duty steps weighted by share of revolutions.)
A4. type (Bearing type) is taken as ball (p = 3). (Deep groove ball bearing.)
A5. F_r (Radial load) is taken as 2.5 kN. (Radial reaction from the shaft calculation.)
A6. F_a (Axial load) is taken as 0 kN. (Locating bearing elsewhere; no axial load here.)
A7. R (Reliability) is taken as 95 %. (Client reliability target.)

A8. a_{iso} (Life modification factor a_{ISO} (from the manufacturer's tool)) is taken as 2. (Read from the manufacturer's life tool for clean ISO VG 68 oil.)
A9. $s_{0,req}$ (Required static safety factor) is taken as 1. (Normal running, smooth loads.)

4. Working

[18] $p = 3$ [1](281:2007 cl. 6–7)
$= 3$
Life exponent, ball bearing

[19] $P = X_1 * F_r + Y_1 * F_a$ [1](281:2007 cl. 6–7)
$= 1 \times 2.5\ kN + 0 \times 0\ kN$
$= 2.5\ kN$
Equivalent dynamic load ($F_a/F_r \le e_{lim}$)

[20] $P_0 = max(X_0 * F_r + Y_0 * F_a, F_r)$ [1](76:2006 cl. 6–7)
$= max(0.6 \times 2.5\ kN + 0.5 \times 0\ kN, 2.5\ kN)$
$= 2.5\ kN$
Equivalent static load

[21] $L_{10} = (C / P)^p$ [1](281:2007 cl. 6–7)
$= (20.3\ kN / 2.5\ kN)^3$
$= 535.39$
Basic rating life, million revolutions

[22] $L_{10h} = L_{10} * 1000000 * 2 * pi / n$
$= 535.39 \times 1000000 \times 2 \times \pi / 1500\ rpm$
$= 5948.7\ h$
Basic rating life in hours

[23] $a_1 = 0.95 * (ln(1 / R) / ln(1 / 0.9))^(2 / 3) + 0.05$ [1](281:2007 cl. 9)
$= 0.95 \times (ln(1 / 95\ \%) / ln(1 / 0.9))^(2 / 3) + 0.05$
$= 0.63791$
Reliability factor (derived, ISO 281:2007 Weibull form)

[24] $L_{nm} = a_1 * a_{iso} * L_{10}$ [1](281:2007 cl. 9)
$= 0.63791 \times 2 \times 535.39$
$= 683.06$
Modified rating life, million revolutions

[25] $L_{nmh} = a_1 * a_{iso} * L_{10h}$
$= 0.63791 \times 2 \times 5948.7\ h$
$= 7589.6\ h$
Modified rating life in hours

[26] $s_0 = C_0 / P_0$ [1](76:2006)
$= 11.2\ kN / 2.5\ kN$
$= 4.48$
Static safety factor

[27] $u_{life} = L_{req} / L_{nmh}$
$= 5000\ h / 7589.6\ h$
$= 0.6588$
Life utilisation (required / modified life)

[28] $u_{s0} = s_{0,req} / s_0$
$= 1 / 4.48$
$= 0.22321$
Static utilisation (required / actual safety factor)

5. Results

SYMBOL	VALUE	DESCRIPTION
L_{nmh}	7590 h	Modified rating life
L_{10h}	5949 h	Basic rating life

SYMBOL	VALUE	DESCRIPTION
P	2.500 kN	Equivalent dynamic load
P_0	2.500 kN	Equivalent static load
s_0	4.48	Static safety factor
L_{10}	535.39	Basic rating life (Mrev)
a_1	0.63791	Reliability factor
L_{nm}	683.06	Modified rating life (Mrev)
u_{life}	0.66	Life utilisation (governing)
u_{s0}	0.22	Static utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Modified rating life against required life	u_{life} = 0.66	u_{life} ≤ 1	0.66	[1](281:2007 cl. 9)
PASS	Static safety factor against required	u_{s0} = 0.22	u_{s0} ≤ 1	0.22	[1](76:2006)

7. Applicability

WARNING records: tabulated values not taken from a reference record are C, C_0 (Bearing load ratings); e_{lim}, X, Y, X_0, Y_0 (Bearing X, Y, e factors); a_{iso} (Bearing a_{ISO}). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
X_1: not entered, so the default of 1 is used.
Y_1: not entered, so the default of 0 is used.

8. Verification

```
Template mech.bearing-life@1.2.0 Content hash 53a676ddb446057a Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. International Organization for Standardization. Rolling Bearings—Dynamic Load Ratings and Rating Life. ISO 281. ISO.
2. ISO 76:2006.
3. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.